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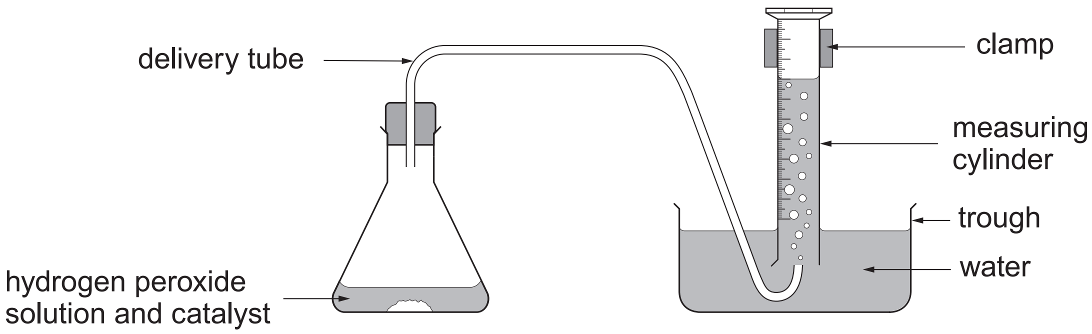
5. Hydrogen peroxide decomposes to give water and oxygen.

(a) Complete the symbol equation to show the reaction taking place. [2]

$$2H_2O_2 \longrightarrow \boxed{}\ H_2O\ +\ \ldots\ldots\ldots\ldots$$

(b) The rate of decomposition of hydrogen peroxide can be measured using the following apparatus.



The rate was investigated using three different catalysts. The results are shown in the table.

Time (s)	Volume of gas collected (cm^3)		
	Catalyst 1	Catalyst 2	Catalyst 3
0	0	0	0
20	2	20	8
40	4	34	15
60	6	38	23
80	8	40	30
100	10	40	36

(i) State which is the **least** effective catalyst. Give a reason for your answer. [1]

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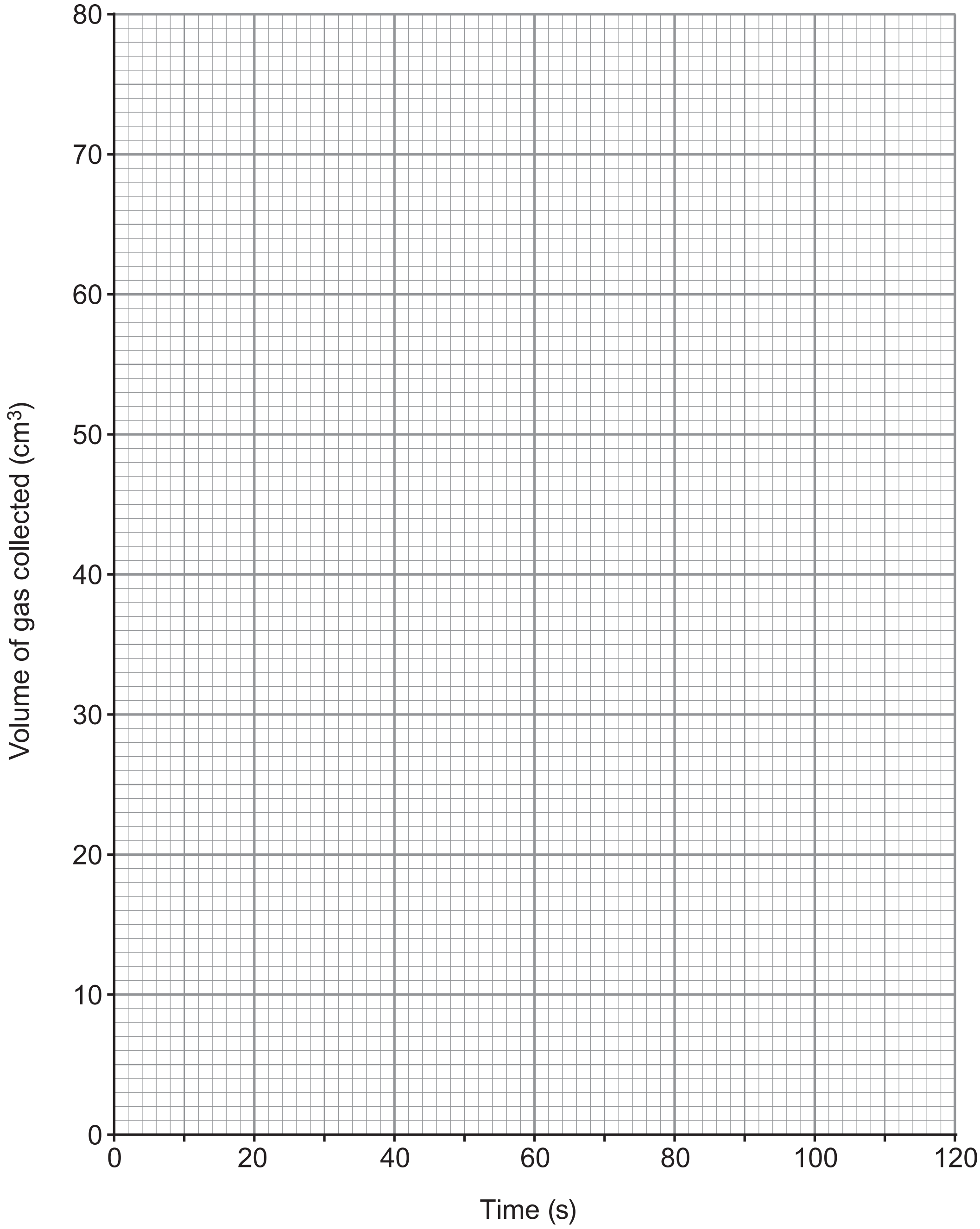
(ii) Plot a graph of the volume of gas collected using **catalyst 2**.
Draw a suitable line. [3]



(iii) **On the same grid**, sketch the graph you would expect to obtain if you added the same amount of catalyst 2 to the same volume of hydrogen peroxide of **twice** the concentration. [2]

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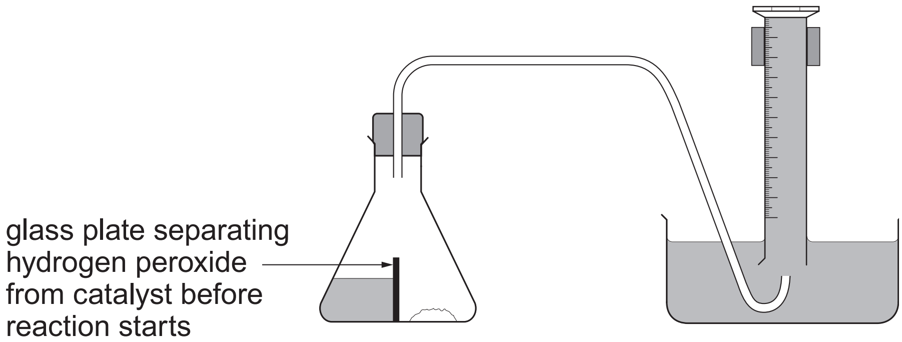
(iv) Another student claimed that he could collect more accurate results using the following apparatus.



Suggest how this apparatus could improve the accuracy of the results. [2]

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